

Independent Replication of *Quantum Counting*
(Brassard, Høyer, Tapp, 1998)
arXiv:quant-ph/9805082

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Abstract

We independently reproduce the principal quantitative claim of Brassard, Høyer and Tapp’s 1998 paper *Quantum Counting*: Theorem 5’s error bound $|t - \tilde{t}| < \frac{2\pi}{P} \sqrt{tN} + \frac{\pi^2}{P^2} N$ with success probability $\geq 8/\pi^2$ for the counting algorithm $\text{COUNT}(F, P)$ that combines a Grover iteration with quantum phase estimation. Two independent Python simulators were built — an exact analytic QPE marginal (via the Dirichlet kernel on the two Grover eigenphases) and a full gate-level Qiskit 2.5 statevector circuit — and cross-validated to $\|\cdot\|_\infty \leq 3 \times 10^{-15}$ across seven diverse (n, t, p) cases. A 90-configuration sweep over $N \in \{16, 32, 64\}$, varying t and counting qubits $p \in \{3, \dots, 8\}$, satisfies Theorem 5’s bound and success-probability guarantee in 90/90 cases. LLM-judge (Argo `argo:gpt-5.4`) verdict: **PARTIAL** (agreement 1.0, coverage 0.67): the counting theorem and its counting-qubit scaling are strongly reproduced; the broader amplitude-amplification claims (Theorems 1–3) were not exercised numerically.

1 Paper summary

The 1998 paper unifies Grover’s algorithm and Shor’s phase estimation into a *quantum counting* algorithm. Given a Boolean oracle $F : \{0, \dots, N-1\} \rightarrow \{0, 1\}$ with t marked inputs and a search space $N = 2^n$, the algorithm $\text{COUNT}(F, P)$ uses $P = 2^p$ counting qubits to apply controlled powers of the Grover operator

$$G = (2|\psi\rangle\langle\psi| - I) O_F, \quad |\psi\rangle = H^{\otimes n}|0\rangle, \quad (1)$$

and inverse quantum Fourier transform on the counting register. The two non-trivial eigenphases of G are $\pm 2\theta$ with $\sin^2 \theta = t/N$; measuring the counting register yields an integer f from which $\tilde{t} = N \sin^2(\pi f/P)$ is computed. The paper proves

$$|t - \tilde{t}| < \frac{2\pi}{P} \sqrt{tN} + \frac{\pi^2}{P^2} N \quad \text{with probability} \geq \frac{8}{\pi^2} \approx 0.811. \quad (2)$$

2 Claims table

ID	Claim	Testable?	Tested?
C1	Amplitude amplification: $\Theta(1/\sqrt{a})$ speedup for any A (Thm 1–3)	Yes (asymptotic)	No
C2	COUNT(F, P) satisfies Eq. (??) with prob. $\geq 8/\pi^2$ (Thm 5/6)	Yes (numerical)	Yes
C3	Corollary 2: $P = c\sqrt{N} \Rightarrow t - \tilde{t} < \frac{2\pi}{c}\sqrt{t} + \frac{\pi^2}{c^2}$	Yes (numerical)	Yes (consis
C4	Heuristic-search quadratic speedup (Thm 4)	Structural / setup-specific	No

3 Method

1. **Paper retrieval.** PDF pulled from <https://arxiv.org/pdf/quant-ph/9805082> (176 410 bytes, 12 pages, PDF 1.4).
2. **Extraction.** `pdftotext -layout` into `work/paper.txt` (581 lines), copied into `extraction/marker.md` and `extraction/nougat.mmd` with header noting the pdftotext substitution (Marker/Nougat not installed locally and paper not in central corpus).
3. **Environment.** Python venv at `work/venv`, Qiskit 2.5.0 + qiskit-aer 0.17.2 + NumPy.
4. **Analytic implementation** (`work/quantum_counting.py`). For each (n, marked, p) we compute the exact marginal distribution over the measured integer f using the standard QPE Dirichlet-kernel form:

$$P(f | \phi) = \frac{\sin^2(\pi P(\phi - f/P))}{P^2 \sin^2(\pi(\phi - f/P))}.$$

The state $|\psi\rangle$ decomposes as an equal superposition of the two eigenvectors of G with eigenphases $\phi_+ = \theta/\pi$ and $\phi_- = 1 - \theta/\pi$, so the counting register’s marginal is $\frac{1}{2}P(f | \phi_+) + \frac{1}{2}P(f | \phi_-)$.

5. **Gate-level cross-check** (`work/verify_qiskit.py`, `work/verify_qiskit_multi.py`). A Qiskit circuit with p counting qubits and n search qubits, initial Hadamards, controlled- G^{2^j} (control = `count_reg[j]`), and inverse QFT with swaps is built. The full statevector is computed and marginalised to the counting register. Convention debugging (see §??) uncovered a subtle Qiskit little-endian issue with the control qubit position in `UnitaryGate.append`: to use the block-diagonal $\text{diag}(I, U)$ form, the control must be the *last* argument (MSB of the gate’s matrix index). After the fix, analytic and gate-level marginals agree to $\|\cdot\|_\infty \leq 3 \times 10^{-15}$ across seven diverse cases.
6. **Sweep** (`work/quantum_counting.py`). Configurations: $n \in \{4, 5, 6\}$, t values per n ($\{1, 3, 6, 10, 15\}$; $\{1, 5, 12, 20, 27, 31\}$; $\{1, 10, 25, 40, 55, 63\}$), and $p \in \{3, 4, 5, 6, 7, 8\}$, giving 90 rows. For each row we record t_{argmax} , $|t - \tilde{t}|$, the Theorem 5 bound, and the *exact* probability that the measurement lands in the “success” set $\{f : |t - N \sin^2(\pi f/P)| < \text{bound}\}$.
7. **LLM-judge** (`work/llm_judge.py`). Free Argo endpoint (`localhost:44497`, `argo:gpt-5.4`). Prompt included the sweep summary, cross-verification log, and paper text; judge returns per-claim verdicts + overall JSON. Raw output at `report/evidence/llm_judge_raw.txt`; parsed at `report/evidence/llm_judge.json`.

4 Results vs paper

4.1 Sanity: gate-level vs analytic (7 cases)

Worst $\|\cdot\|_\infty$ deviation: 2.72×10^{-15} . All argmax pairs are either identical or mirror-symmetric (f vs $P - f$), which is expected because QPE assigns equal probability to $\phi_+ = \theta/\pi$ and $\phi_- = 1 - \theta/\pi$, and both map to the same $\tilde{t} = N \sin^2(\pi f/P)$. See `report/evidence/qiskit_verify_multi.log`.

4.2 Main sweep (90 configs)

Every one of the 90 configurations satisfied both the Theorem 5 error bound and the $8/\pi^2$ success-probability guarantee:

Argmax within Theorem-5 bound	90 / 90
Exact $P(\text{within bound}) \geq 8/\pi^2$	90 / 90

Representative rows (t = true count, $\tilde{t}$ = argmax estimate, B = Thm-5 bound, p_s = exact success prob):

n	t	p	$\tilde{t}$	$ t - \tilde{t} $	B	p_s
4	1	8	1.039	0.039	0.101	1.000
4	3	6	2.925	0.075	0.719	0.996
5	12	6	11.36	0.645	2.00	1.000
5	20	7	19.89	0.112	1.26	1.000
6	25	8	24.99	0.011	0.99	1.000
6	40	5	38.24	1.76	10.55	1.000
6	63	4	61.56	1.44	27.40	1.000

The error monotonically shrinks with P as $\sim P^{-1}$ in the leading term, consistent with Corollary 2.

5 Verdict

PARTIAL (with agreement = 1.00 on the tested claims, coverage = 0.67). Justification: the paper’s principal quantitative counting result (C2/Theorem 5) is reproduced independently at machine precision, in every one of 90 test configurations across three search-space sizes. Corollary 2’s scaling (C3) is consistent with our sweep. The general amplitude-amplification speedup claims (C1) and heuristic-search theorem (C4) were not exercised numerically in this replication and would require additional benchmarks with families of algorithms A of varying success probability a .

6 What worked and what didn’t

Worked

- Analytic QPE marginal derivation (Dirichlet kernel over the two Grover eigenphases) — gave a closed-form target the gate-level implementation could be checked against, avoiding a fully-blind reproduction.
- Exact success-probability computation — we sum the marginal over the theorem’s success set instead of relying on shot statistics, giving the crispest possible check of the $8/\pi^2$ bound.
- Cross-validation to 10^{-15} ruled out both circuit and analytic bugs.

Didn't work at first

- First Qiskit gate-level attempt disagreed with the analytic form by $\|\cdot\|_\infty \approx 0.6$ and returned nonsense $\tilde{t}$ values. Root cause: with the standard block form $\text{full} = \text{diag}(I, U)$ passed to `UnitaryGate`, the control qubit must be the *last* argument in `qc.append(cU, [..., control])` in Qiskit's little-endian convention (MSB of the gate matrix index). Placing the control as the first argument silently applied U to a wrong superposition of control states. Fix: swap to `list(search_reg) + [count_reg[j]]`.
- Argo `claude-opus-4.8` judge call failed with an upstream JSON parse error on the large prompt. Fell back to `argo:gpt-5.4`, which returned a well-formed judgement.

Caveats

- The sweep is restricted to $n \leq 6$ (dense-matrix G powers become expensive beyond that). The theorem is asymptotic in nature; our agreement in the small- N regime is a necessary but not fully exhaustive test.
- Corollary 2's exact scaling $P = c\sqrt{N}$ was not *fit* parametrically; we only verified that shrinking P -dependent bound held pointwise.

7 Open Questions

- Q1. Tightness of the $8/\pi^2$ probability floor for intermediate t/N .** The exact success probability in our sweep was typically ≥ 0.99 across all (n, t, p) , well above the theoretical ≈ 0.811 . Where does the paper's bound bite? For which t/N and P is the marginal actually near the floor?
- Q2. Behaviour when $t/N \rightarrow 0$ or $t/N \rightarrow 1$.** The eigenphase $\theta \rightarrow 0$ regime gives progressively worse leading-term error $\propto \sqrt{tN}/P$; empirically we saw the argmax hop to nearby f but remained within the bound. What is the actual worst-case t/N for a fixed P ?
- Q3. Bias of $\tilde{t}_{\text{argmax}}$ vs the paper's "round" rule.** The paper does not commit to argmax; it just bounds any output of the algorithm. In practice *which* decoding rule (argmax, weighted median, matched-filter over the two peaks) is best for a fixed query budget?
- Q4. Non-uniform initial states.** The whole analysis assumes the search register starts in the exact uniform superposition $A|0\rangle = H^{\otimes n}|0\rangle$. What happens under imperfect A (e.g. depolarising or dephasing noise on the counting or search register), which is inevitable on real hardware?
- Q5. Comparison to iterative-QPE / Bayesian counting.** Modern variants (Grinko-et-al. iterative amplitude estimation; Bayesian phase estimation) claim comparable or better sample complexity with fewer counting qubits. Under our exact framework, at what N, t does one method dominate?